

COMPUTER SYSTEM REFURBISHMENT

CSACR401

Refurbish Computer System

Competence

RQF Level: 4

Learning Hours

70

Credits: 7



Sector: ICT and Multimedia

Trade: Computer System and Architecture

Module Type: Specific

Curriculum: ICTCSA4001- TVET Certificate IV Computer System and Architecture

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Purpose statement	This module describes the skills, knowledge and attitude required to refurbish computer system. This module is intended to prepare learners pursuing TVET Level 4 in Computer Systems and Architecture. At the end of this module the learners will be able to perform computer system renovation, restoration and conversion.					
Learning assumed to be in place	▪ Computer system maintenance ▪ Computer peripherals deployment ▪ Computer system deployment ▪ PCB and Computer System Assembling ▪ Implement basics of electronic circuits					
Delivery modality	Training delivery		100%	Assessment		Total 100%
	Theoretical content		30%	Formative assessment	30%	50%
	Practical work:		70%		70%	
	● Group project and presentation	20%				
	● Individual project /Work	50%				
			Summative Assessment		50%	

Elements of Competence and Performance Criteria

Element of competence	Performance criteria
1.Perform Computer System Renovation	1.1. Working Place is Properly Prepared According to occupational safety and health standards
	1.2. Computer system components are properly selected based on system requirement
	1.3. Tools, materials and equipment are properly selected based on system requirement
	1.4. Computer system is correctly disassembled based on its architecture
	1.5. New hardware components are properly installed based on computer system requirement
	1.6. Computer system is correctly assembled based on renovation requirement
	1.7. system software are properly installed based on computer system requirement
	1.8. Testing is systematically performed according to computer unit testing requirement
2.Perform Computer System Restoration	2.1. Computer system motherboard level troubleshoot is correctly performed based on computer system functionality
	2.2. Computer system software troubleshooting is correctly performed based on computer system functionality
	2.3. System testing is correctly performed according to computer system restoration requirement

3. Perform Computer System Conversion	3.1. Transformation requirement are properly identified based repurposing goals	
	3.2. Computer system transformation techniques are properly applied based on intended use	
	3.3. New computer system is properly assembled based on repurposing goals	
	3.4. System testing is successfully performed based on system expected functionality	
	3.5. Refurbishment report is effectively elaborated based on work done	
Intended Knowledge, Skills and Attitude		
Knowledge	Skills	Attitude
✓ Identify computer system components	✓ Set a working place	✓ Innovative
✓ Describe of assemble and disassemble Computer system	✓ Select a computer system component	✓ Organization
✓ Identify computer system drivers and software	✓ Assemble and disassembling computer components	✓ Safety First
✓ Describe computer system transformation	✓ Install a new hardware component	✓ Flexibility
	✓ Upgrade a computer hardware component	✓ Time Management
	✓ Troubleshoot a system software	✓ Problem Solving
	✓ Test a full functionality of computer system	✓ Forecasting
	✓ Elaborate a computer system report	✓ Team work
		✓ Hard worker
		✓ Self-motivated

techniques		
Course content		
Learning outcomes	At the end of the module the learner will be able to: 1. Perform computer system renovation 2. Perform computer system restoration 3. Perform computer system conversion	
Learning outcome 1: Perform computer system renovation	Learning hours: 20	
Indicative content		
• Setting Working Place ✓ Identify Hazards ✓ Implement Safety Measures ✚ Provide Personal Protective Equipment (PPE) ✚ Ergonomics ✚ Electrical Safety ✚ Fire Safety ✚ Hygiene and Sanitation • Selection of Computer system components ✓ Hardware ✚ Central processing unit ✚ Motherboard ✚ Storage and memories ✚ Input device/Output device ✚ Adapter cards ✚ Power Supply Unit (PSU)		

✓ Software

 System software

 Application software

• **Selection of Tools, materials and equipment**

✓ Tools

 ESD TOOLS

 Hand tools

 Diagnostic tools

 Cleaning tools

✓ Materials

 Thermal paste

 Cleaning materials

 Adhesive materials

✓ Equipment

 Power protection devices

 PPEs

• **Disassembling Computer system**

✓ Description of Computer system

 Type

 Categories

 Components

✓ Implementation of Computer system disassembling procedures.

• **Installation of New hardware component**

✓ Identification of the new hardware component

✓ Check hardware component compatibility

✓ Implementation of new hardware installation procedures

- **Assembling Computer system**
 - ✓ Implementation of Computer system assembling procedures
 - ✓ Perform appearance customization.
- **Installation of Computer system drivers and software**
 - ✓ Perform software compatibilities checking.
 - ✓ Implementation of software installation procedures
 -  System software
 -  Application software
- **Perform computer system test**
 - ✓ System functionality
 - ✓ System performance
 - ✓ System stress test

Resources required for the learning outcome

Equipment	▪ Power supply tester ▪ Oscilloscope ▪ Multi-meter ▪ Function generator ▪ Spectrum analyser ▪ Power supply ▪ Blower
Materials	▪ Thermal paste ▪ Isopropyl alcohol and cotton swabs ▪ Electrical tape and zip ties ▪ Spare screws ▪ Labelling materials
Tools	▪ Screwdrivers ▪ Needle-nose pliers ▪ Tweezers

	▪ Flashlight or headlamp ▪ Cable testers ▪ Thermal paste applicator ▪ Wire cutter/strippe ▪ Pliers ▪ ESD Tools
Facilitation techniques	▪ Demonstration and simulation ▪ Individual and group work ▪ Trainer guided ▪ Group discussion ▪ Practical exercise
Formative assessment methods	▪ Product assessment ▪ Written assessment ▪ Performance assessment ▪ Oral presentation

Learning outcome 2: Perform computer system restoration	Learning hours: 30
Indicative content	
• Perform Computer system motherboard troubleshooting ✓ Description of computer system motherboard + Definition of computer system motherboard + Main components of computer system motherboard and their function + Interconnections and Communication of computer system motherboard components + Computer system motherboard technology	

- ✓ Identification of the common faults.
- ✓ Perform computer system unit testing.
- ✓ Perform repair computer system motherboard.
 -  Trace repair
 -  Soldering and desoldering
 -  Component board level repair
- ✓ Perform testing of the computer system motherboard component
- ✓ Assembling computer system motherboard
- **Perform Computer system software troubleshooting**
 - ✓ Identification of common faults
 - ✓ Perform data recovery.
 - ✓ Perform factory reset.
 - ✓ Restoration of operating system
 - ✓ Perform firmware upgrade.
 - ✓ Perform Computer system unit testing
- **Perform computer system testing**
 - ✓ Hardware testing
 - ✓ Software testing

Resources required for the indicative content

Equipment	▪ Power supply tester ▪ Oscilloscope ▪ Multi-meter ▪ Function generator ▪ Workbench or anti-static mat ▪ SMD rework station ▪ Blower
Materials	▪ Anti-static wrist strap/mat ▪ Thermal paste

	▪ Isopropyl alcohol and cotton swabs ▪ Electrical tape and zip ties ▪ Spare screws ▪ Labelling materials ▪ Soldering wire (tin) ▪ Soldering paste ▪ Desoldering wick ▪ Glue stick
Tools	▪ Screwdrivers ▪ Needle-nose pliers ▪ Tweezers ▪ Flashlight or headlamp ▪ Cable testers ▪ Thermal paste applicator ▪ Wire cutter/striper ▪ Pliers ▪ Soldering iron ▪ Desoldering pump ▪ Glue gun
Facilitation techniques	▪ Demonstration and simulation ▪ Individual and group work ▪ Practical exercise ▪ Trainer guided ▪ Group discussion
Formative assessment methods	▪ Product assessment ▪ Written assessment ▪ Performance assessment ▪ Oral presentation

Learning outcome 3: Perform Computer System Conversion

Learning hours: 20

Indicative content

- **Identification of computer system transformation requirement**
 - ✓ Description of computer system transformation
 - + Definition
 - + Purpose
 - + Computer system components technologies
 - ✓ Identification of computer system components
- **Applying computer system transformation techniques.**
 - ✓ Identification of Computer system transformation techniques
 - + Software based
 - + Hardware based
 - + Networking and connectivity
 - + Hybrid
 - ✓ Use Computer system transformation techniques.
- **Assembling new computer system**
 - ✓ Apply assembling precaution.
 - ✓ Apply assembling procedure.
- **Perform computer System test**
 - ✓ System Compatibilities
 - ✓ System performance
 - ✓ System functionality
- **Elaboration of System report**
 - ✓ Status before maintenance

✓ Activity procedures ✓ Status after maintenance ✓ Conclusion and recommendation	
Resources required for the indicative content	
Equipment	▪ Power supply tester ▪ Oscilloscope ▪ Multi-meter ▪ Function generator ▪ Workbench or anti-static mat ▪ SMD rework station
Materials	▪ Anti-static wrist strap/mat ▪ Thermal paste ▪ Isopropyl alcohol and cotton swabs ▪ Electrical tape and zip ties ▪ Spare screws ▪ Labelling materials ▪ Soldering wire (tin) ▪ Soldering paste ▪ Desoldering wick ▪ Glue stick
Tools	▪ Screwdrivers ▪ Needle-nose pliers ▪ Tweezers ▪ Flashlight or headlamp ▪ Cable testers ▪ Thermal paste applicator ▪ Wire cutter/striper

	▪ Pliers ▪ Blower ▪ Soldering iron ▪ Desoldering pump ▪ Glue gun
Facilitation techniques	▪ Demonstration and simulation ▪ Individual and group work ▪ Trainer guided ▪ Group discussion
Formative assessment methods	▪ Written assessment ▪ Performance assessment ▪ Oral presentation

Integrated/Summative assessment

Integrated situation

ZXY is a company that refurbishes Computer systems hardware in order to return them back on the market. It is located in Kigali city. This company bought second hand electronic equipment from auctions of various institutions. Some of them are functional and others are non-functional, moreover they have various brand, model, and specifications.

The majority of them are computer system hardware components such as CPUs, Motherboards, storage devices, peripherals devices and adapter cards.

As computer system refurbishment technician, you are Hired by **ZXY** Company and required to build a computer with these specifications: CPU of 2.16HZ, 4GB RAM, 500GB HDD, screen 15.6 inch, black cover and generate report for work done.

Tools, materials, and equipment's are provided by **ZXY** company.

This work must be done within 8 hours.

Resources

Tools	▪ Screwdrivers ▪ Needle-nose pliers ▪ Tweezers ▪ Flashlight or headlamp ▪ Cable testers ▪ Thermal paste applicator ▪ Wire cutter/striper ▪ Pliers ▪ Blower ▪ Soldering iron ▪ Desoldering pump ▪ Glue gun ▪ Power tool ▪ hand tool ▪ measuring tool ▪ marking tool ▪ painting tool ▪ cleaning tools ▪ Operating System image ▪ Flash disk
Equipment	▪ PPEs ▪ Variable DC power supply ▪ Benchwork table ▪ Oscilloscope ▪ Function generator ▪ Digital multimeter
Materials/ Consumables	▪ Thermal paste ▪ Cable ties ▪ Cleaning wipers/cloth ▪ Isopropyl alcohol ▪ Cleaning solution ▪ Adhesive tape

		▪ Processor ▪ RAM ▪ HDD ▪ Computer screens ▪ Computer case			
Assessable outcomes	Assessment criteria (Based on performance criteria)	Indicator	Observation		Marks allocation
			Yes	No	
Perform computer system renovation (60%)	Preparation of Working Place	Safety Measures is applied			5
	Selection of Computer system component	CPU of 2.16HZ is selected			3
		4GB RAM is selected			3
		500 HDD is selected			3
		screen 15.6 inch selected			3
		Operating system is selected			3
	Selection of Tools, materials and equipment	Hand tools are selected			3
		Cleaning tools are selected			3
		ESD tools are selected			4
		Painting tools are selected			3
		Diagnostic tools are selected			3
		Thermal paste is			2

		selected			
		Cleaning material are selected			2
		Adhesive materials are selected			3
		Equipment are selected			3
	Installation of New hardware component	hardware component is fixed			2
	Assembling Computer system	Computer system assembling procedures is applied			2
		Appearance customization is performed			3
	Installation of system software	Operating system is installed			4
	Performing test	Computer functionality is performed			4
Perform computer system restoration (20%)	Perform Computer system motherboard level troubleshoot	computer unit test is performed			4
		computer motherboard is repaired.			6
		computer motherboard is assembled			3
	Perform Computer	operating system is well function			3

	system software troubleshooting	computer system components technologies are used			4
Perform computer system conversion (20%)	Assembling New computer system	assembling precaution are applied			3
		CPU of 2.16HZ is installed			3
		4GB RAM is installed			3
		500 HDD is installed			3
		screen 15.6 inch installed			2
		power supply 19V/3A installed			2
	Elaboration of Refurbishment report	Refurbishment report is well done			4
Total marks		100			
Percentage Weightage		100%			
Minimum Passing line % (Aggregate): 70%					

References:

1. Elahi, A. (2006). *Computer Systems*. rondon: springer.
2. Lee, G. (2021). *Troubleshooting & Fixing Common Problems in PCs & Laptops*. Seoul.
3. Mano, M. M. (2007). *Computer System Architecture / Third Edition*. New York: Pearson.
4. Mohammed, A. (2010). *Ultimate Computer Repair Guide*. washington: Academia.
5. Smoot, A. M. (2015). *computer repair business*. Boston: kindle edition.